

UAV Target Detection & Air Quality Monitoring



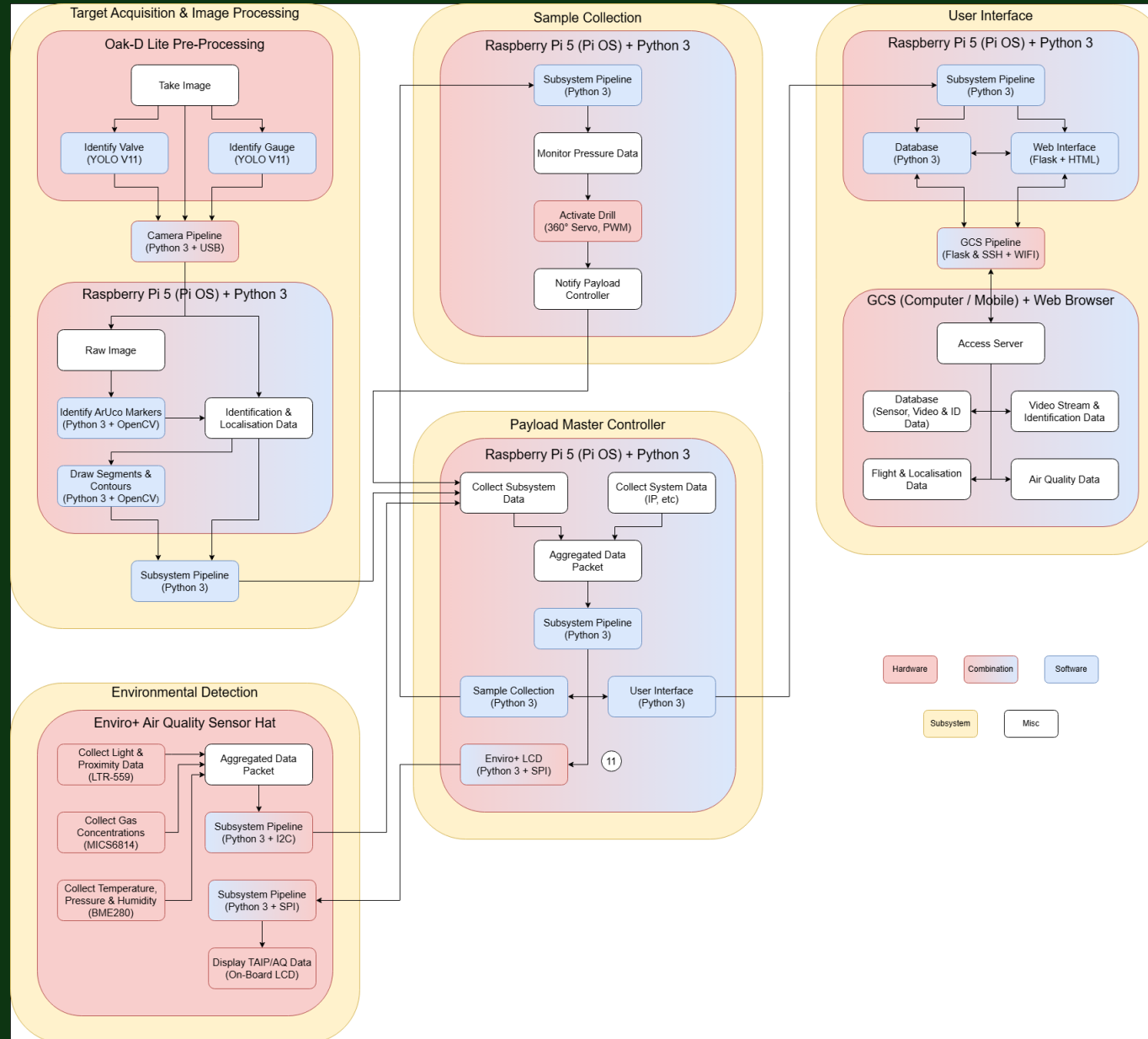
GROUP 24

**EGH455 ADVANCE
SYSTEM DESIGN**

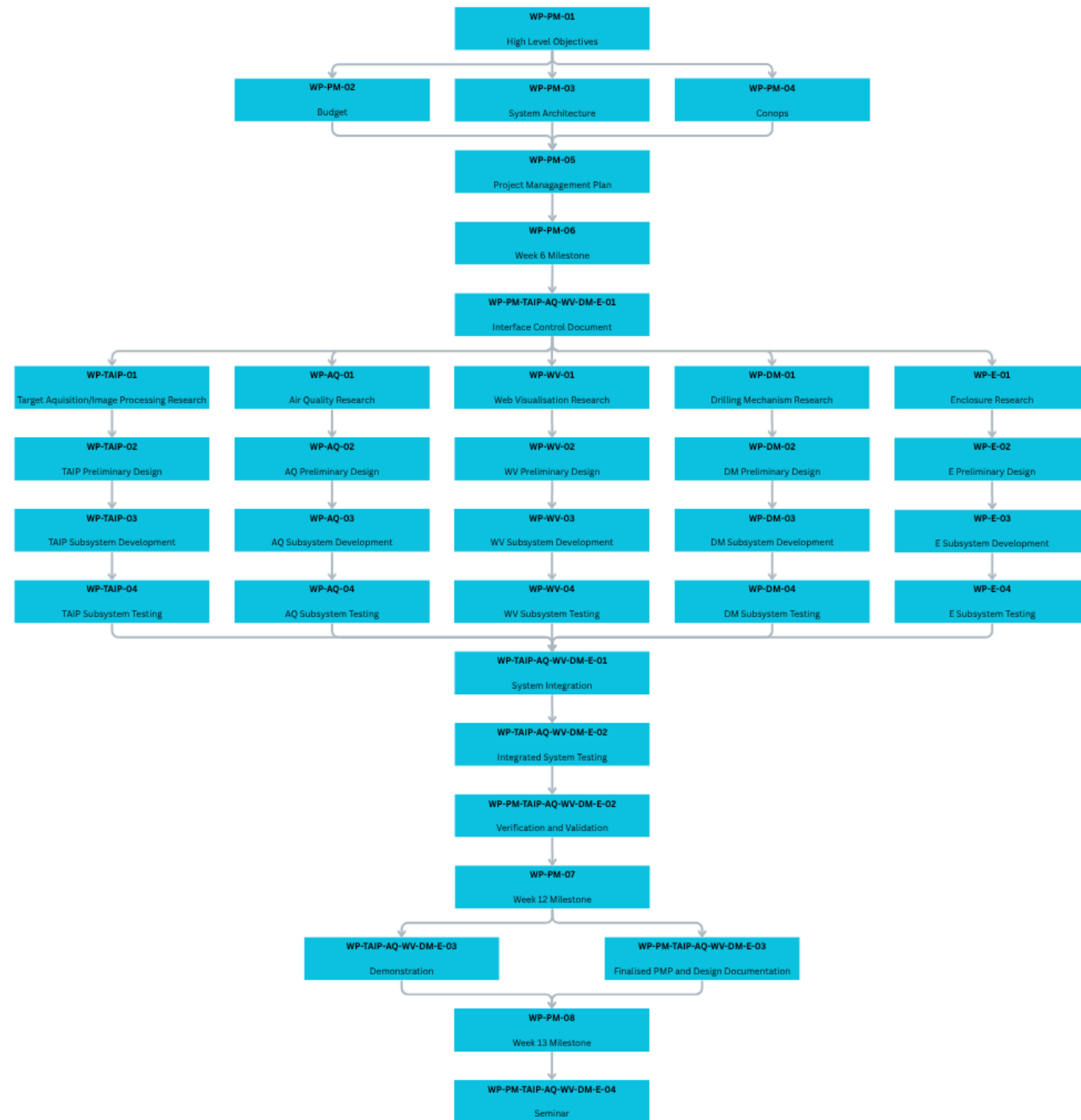
HLO-M-7 System Engineering Methodology



System Architecture

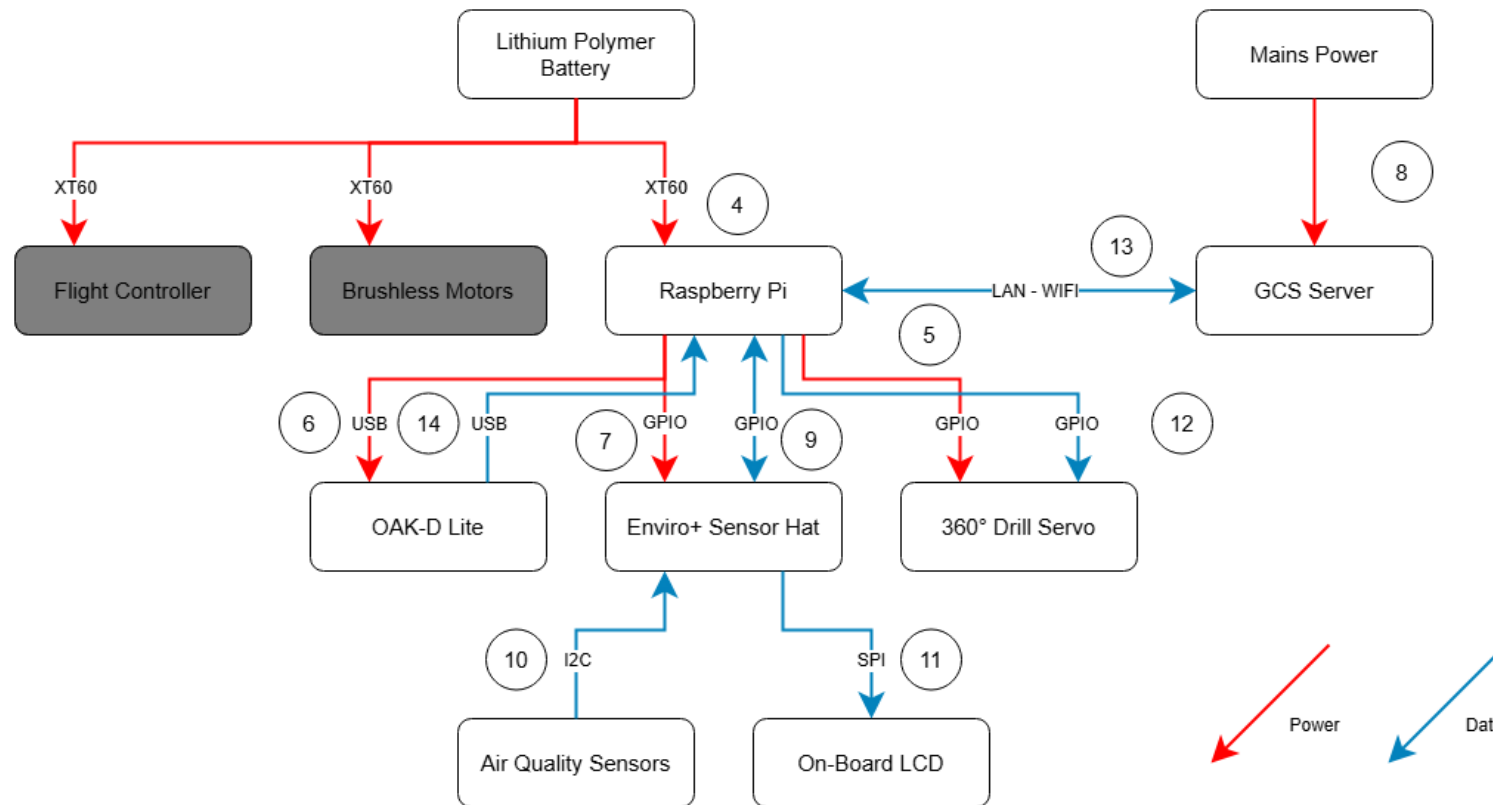


WBS

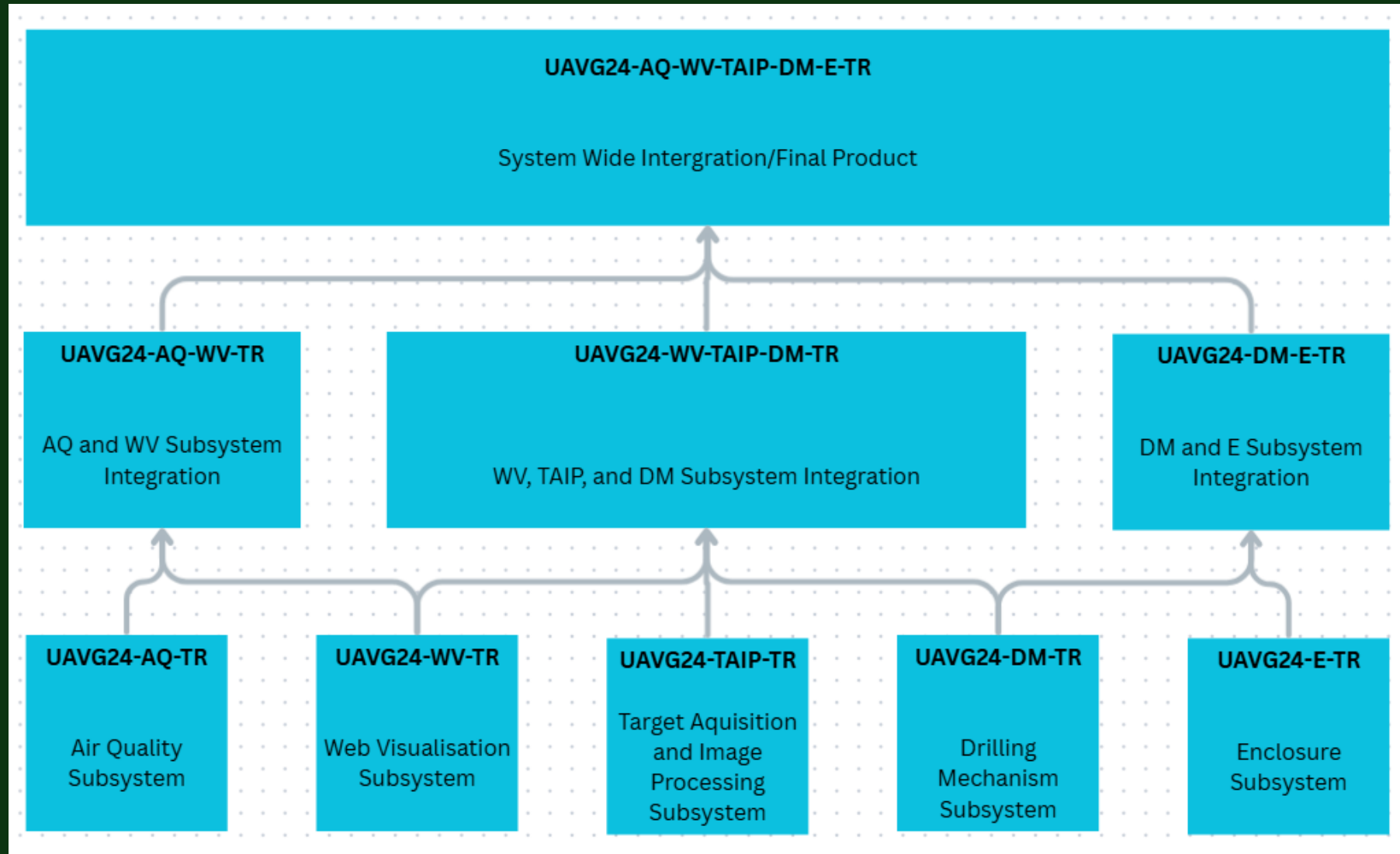


Interface Control Document

System Architecture - Power Distribution & Dataflow



Test Plan



Budget and Timeline

Subsystem	Components	Cost (\$)	
		Estimated	Actual
AQ	Enviro + Air Quality	\$129.95	\$125
DM	DF15RSMG 360 Degree Motor	\$23.95	\$35
DM	Sampler Drill Piece	\$7.18	\$1.69
E	3D Printed Enclosure	\$20	\$1.76
E	Raspberry PI 5 + Active Cooler + PCB	\$234.45	\$120
E	Cables + Accessories	\$56.59	30
TAIP	OAKD-Lite	\$149	\$290
WV	Laptop/Smartphone (Already supplied)	\$0	\$0
Total		\$644.07	\$603.45
Contingency		20%	-
Total (Contingency Included)		\$772.88	\$724.14

Project's Total Lifecycle
13 Weeks

GROUP 24 - UAVPayloadTAQ

**AJAY
ARNOLD
ANBU
ANANTHAN**

**4th Year Bachelor of
Engineering Mechatronics
(Honours)**

Enclosure & Drilling Mechanism



OUR TEAM

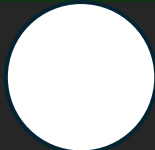
Enclosure and Payload Scope

PAYLOAD ENCLOSURE (PE)

- 1. Provides structural protection and precision mounting for payload components**
- 2. Houses Raspberry Pi 5, OAK-D Lite, Enviro+, Custom PCB and Drilling Mechanism**
- 3. Designed for *IP47* compliance (dust and splash protection)**
- 4. Ensures optimal sensor exposure and easy component access**

DRILLING MECHANISM (DM)

- 1. Custom Rotary Drill System for simulated soil sampling**
- 2. Creates a 15mm x 15mm hole in kinetic sand**
- 3. Controlled via a 360 servo motor for accurate actuation**
- 4. Includes pressure monitoring (<2 bar) to prevent over force and protect system**



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System Requirements

Soil Sampling Simulation (HLO-M-6)

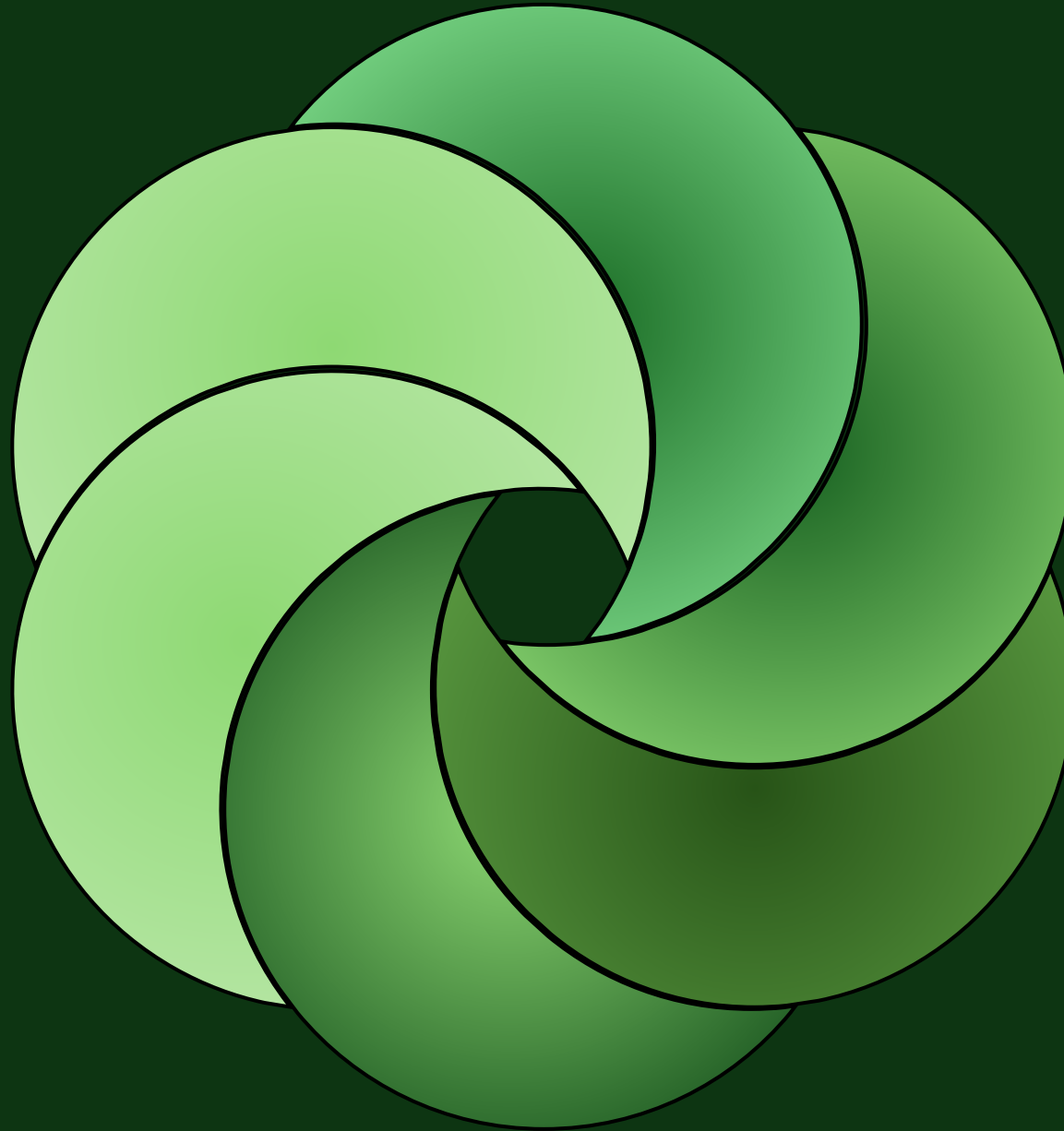
3D printed drill creates
15mm x 15mm hole in
kinetic sand

Conditional Activation (HLO-M-2, HLO-M-6, REQ-F-09)

Drill only operates post
landing under <2 bar
pressure

Weight Compliance (HLO-M-4, REQ-P-01)

Total Payload <410g



Environmental & Structural Protection (HLO-M-4, REQ-P-01)

Enclosure provides IP41
protection and secure
component mounting

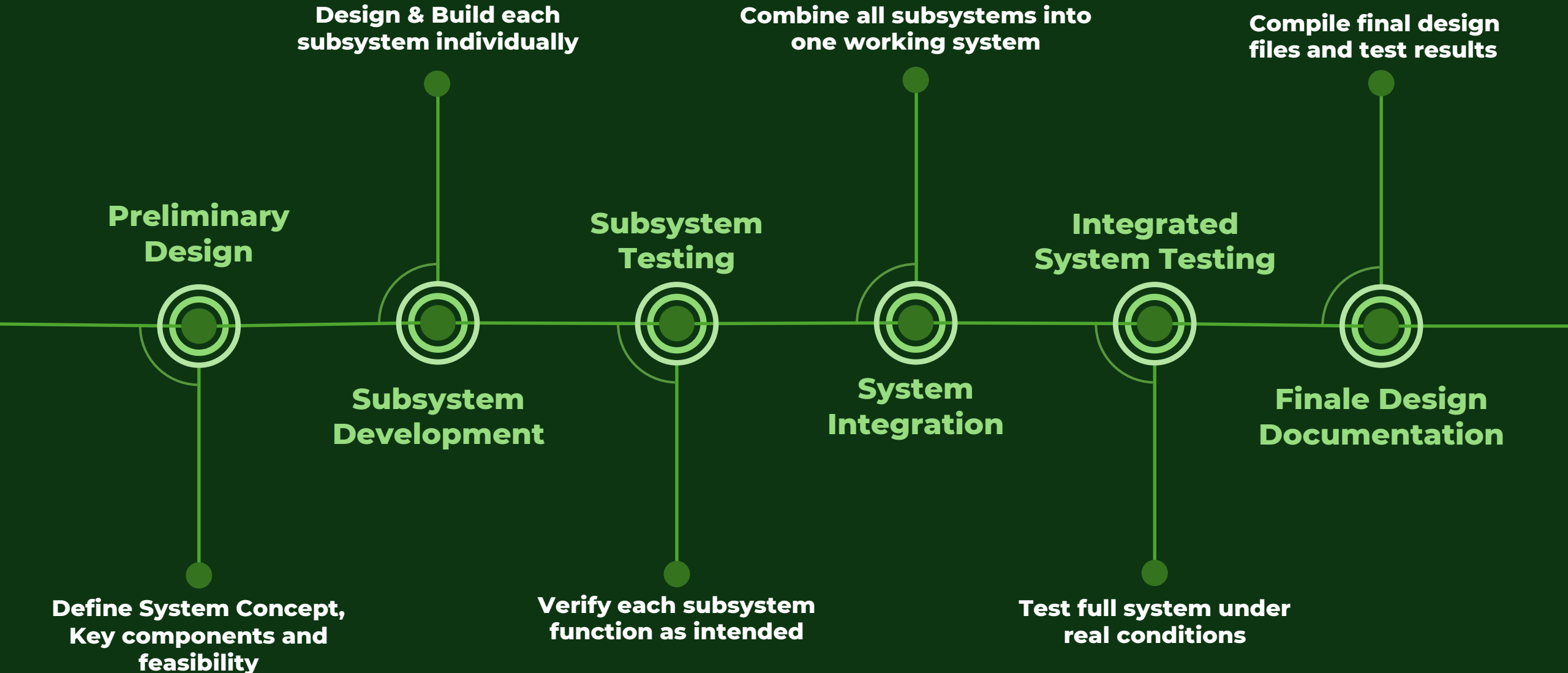
Autonomous Operation & Feedback (HLO-M-4, HLO-M-5)

Payload functions
independently; drill status
displayed on Enviro+ LCD
and sent to GCS

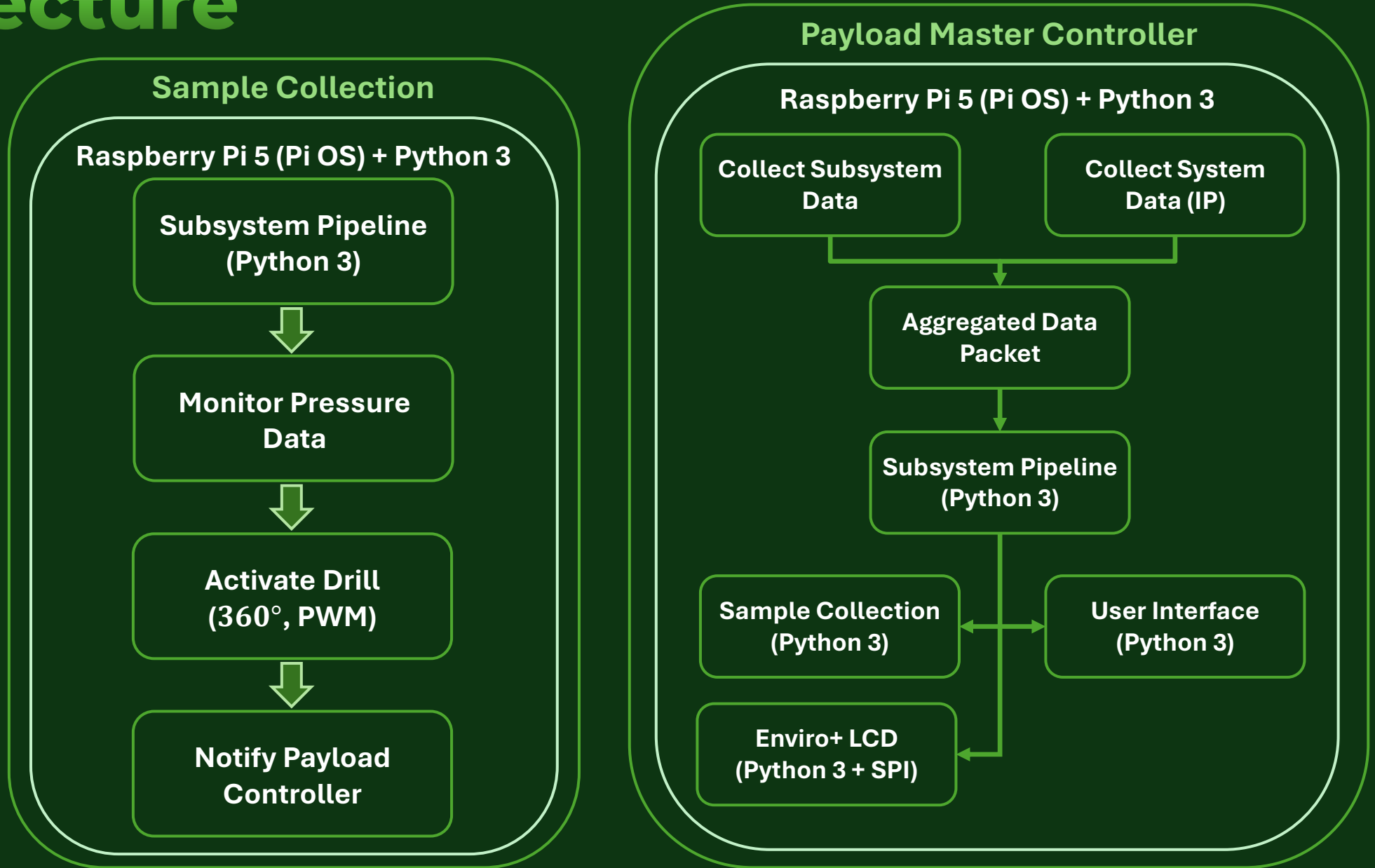
Integration & Compliance (HLO-M-4, HLO-M-7)

Mechanism fits securely
within enclosure;
developed per system
engineering standards

Work Packages



System Architecture Flow



Hardware Design Features

01

Payload Enclosure Components

- Raspberry Pi 5 (Main controller)
- Custom PCB for power distribution
- Enviro+ environmental sensor
- OAK-D Lite stereo camera
- DM integration bay

02

Drilling Mechanism Components

- DF15RSMG 360 continuous servo
- Bevel/ Spur Gear system
- Pinion and Rack system
- Drill mechanism Slider/ Mount
- PLA Drill bit (15mm x 15mm)
- Pressure sensing activation logic

114g

Enclosure + Electronics

110g

Drilling Mechanism

424g

Total Payload Weight

Hardware Design Features

01

Payload Enclosure Components

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298g

Enclosure + Electronics

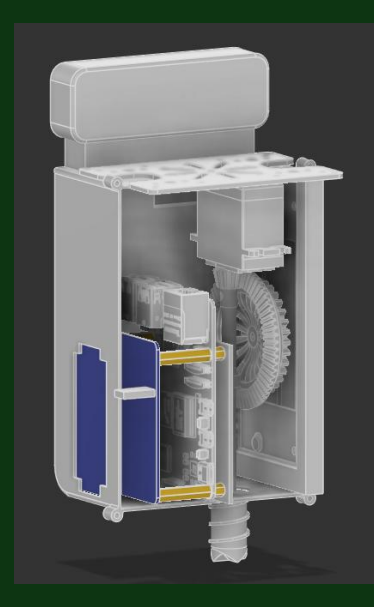
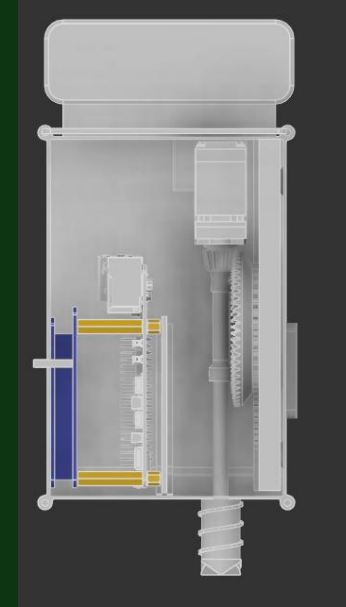
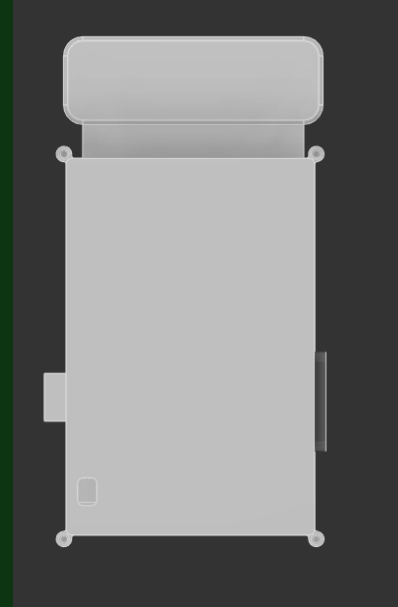
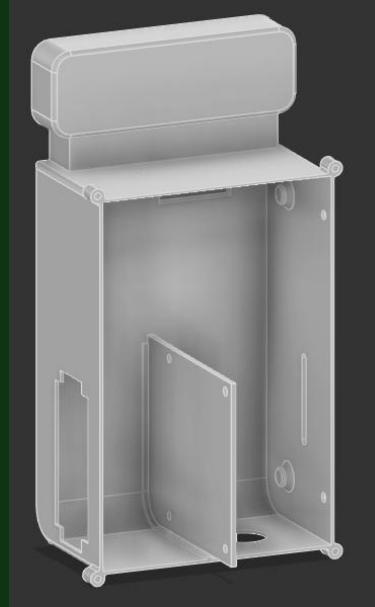
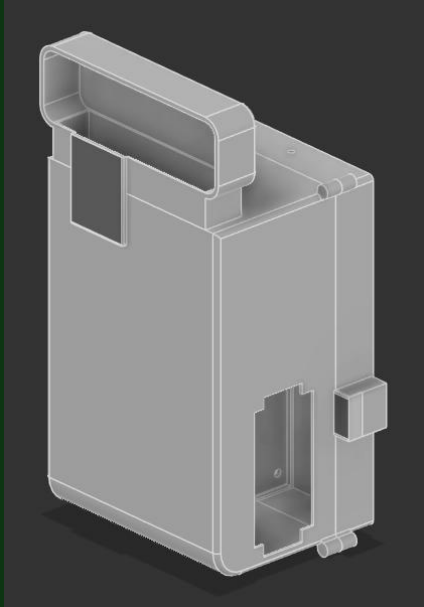
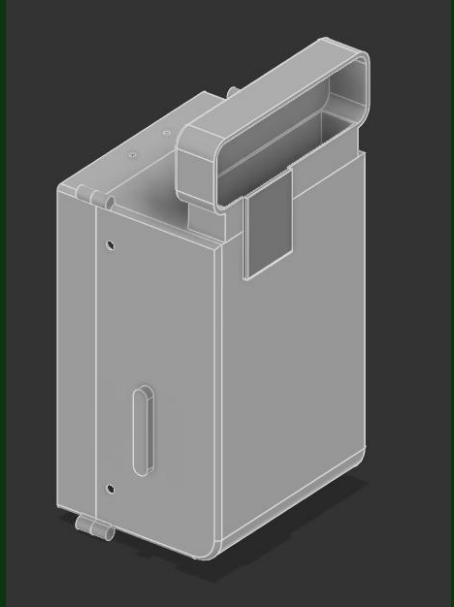
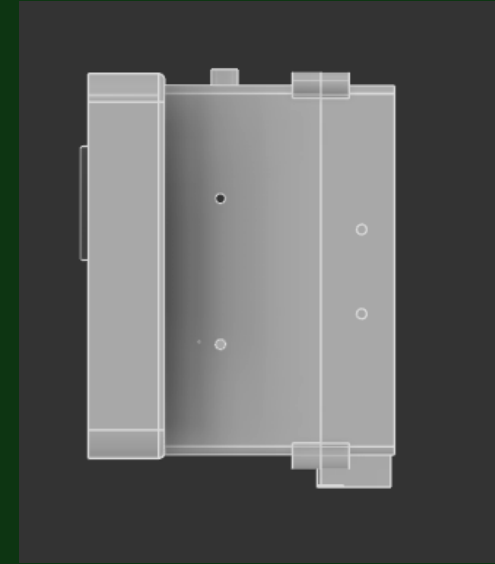
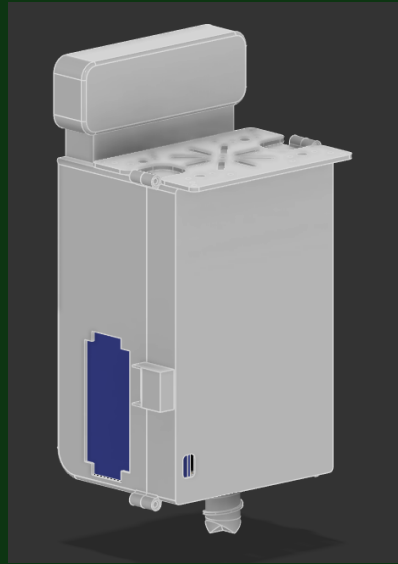
110g

Drilling Mechanism

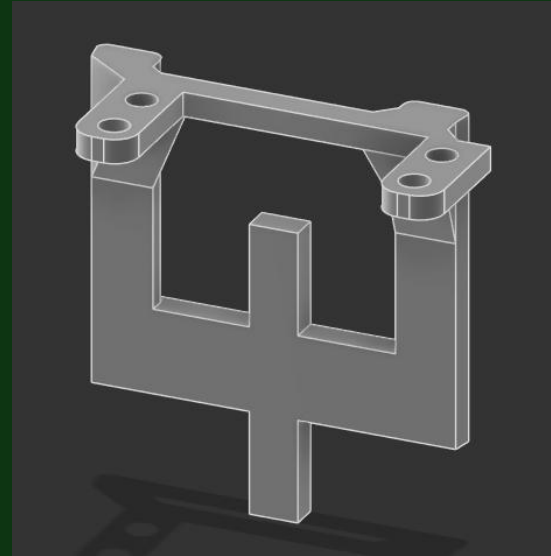
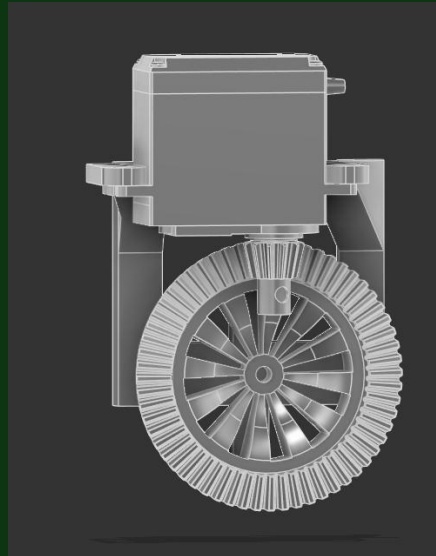
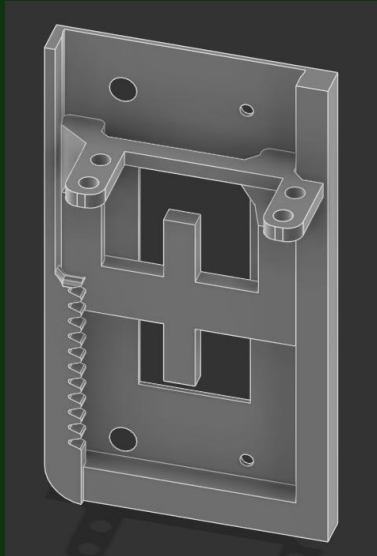
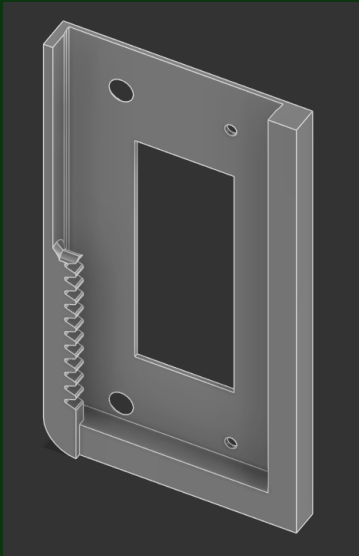
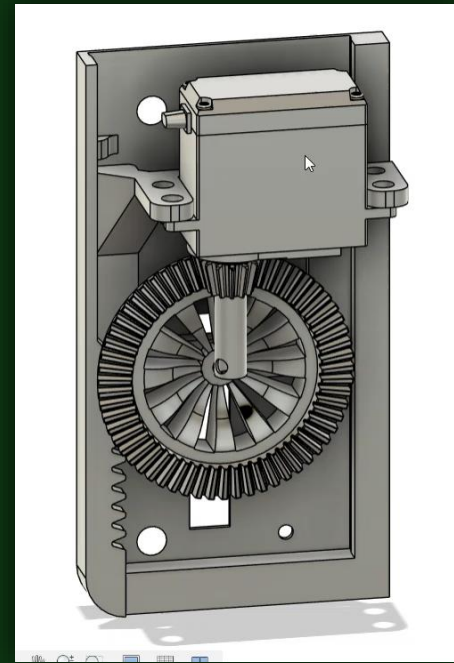
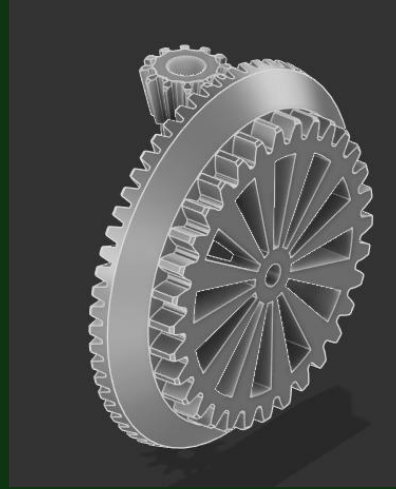
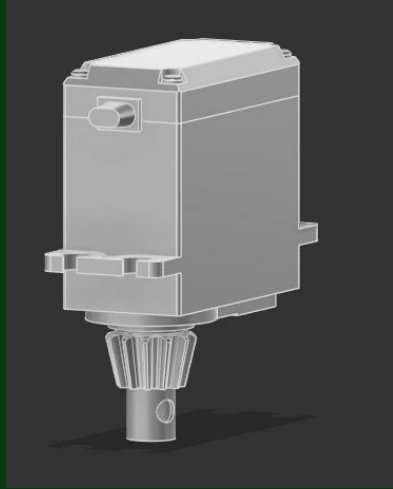
408g

Total Payload Weight

Hardware Design: Payload Enclosure Details



Hardware Design: Drilling Mechanism Details



Software Design: Control Logic & Integration

```
from gpiozero import Servo
from time import sleep
```

```
Initialize servo as None globally
servo = None
```

```
def drill_cycle():
    global servo
    try:
        if servo is None:
            # Use default software PWM (no pigpio
            # needed)
            servo = Servo(13, min_pulse_width=1/1000,
            max_pulse_width=2/1000)
            print("Servo initialized with software
            PWM on GPIO 13")
```

```
        print("Testing servo - moving clockwise")
        servo.value = 1.0
        sleep(1.1)
```

```
        print("Stopping servo")
        servo.value = 0
        sleep(1)
```

```
        print("Moving counter-clockwise")
        servo.value = -1.0
        sleep(1.1)
```

```
        print("Stopping servo")
        servo.value = 0
```

```
        print("Drill cycle complete")
```

```
    except Exception as e:
        print(f"Error in drill cycle: {e}")
        import traceback
        traceback.print_exc()
```

```
if name == "main":
    drill_cycle()
```

Servo Motor Code

```
@app.route("/gauge_values")
def gauge_values():
    """API endpoint for gauge readings"""
    try:
        values = oak_camera.get_gauge_values()

        # FIX: Only log if values exist
        if values and len(values) > 0:
            log_target("gauge", values[0])

        if values and len(values) > 0 and values[0] <= 2:
            try:
                print("Starting drill cycle...")
                success = drill_cycle()

                if success:
                    return jsonify({"status": "drill cycle
                    complete"})
                else:
                    return jsonify({"error": "drill cycle
                    failed"}), 500
            except Exception as e:
                print(f"Servo test error: {e}")
                import traceback
                traceback.print_exc()
                return jsonify({"error": str(e)}), 500
```

Integration Code

Test Report Overview (Enclosure)

ENC-01

Verify total
payload weight is
under 410g



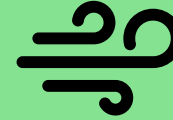
ENC-02

Check enclosure
fits UAV mount
securely with no
obstruction



ENC-03

Ensure Air Quality
sensor is properly
exposed and
readings accurate



ENC-04

Test enclosure for IP41
protection against water and
debris



ENC-05

Confirm structural
durability under
drop and vibration



ENC-06

Validate LCD visibility and
port accessibility in all
lighting



ENC-07

Check for safe internal
temperature during operation

DRL-01

Check Drill Only
activates when
pressure <2 bar

DRL-02

Verify drilled hole meets
15mm x 15 mm size
tolerance

DRL-03

Ensure drill
retracts fully after
operation

DRL-04

Test Consistent
drilling and
retraction over
multiple runs

DRL-05

Validate accurate
servo motion and
control response

DRL-08

Confirm drill
integrates
smoothly with
enclosure
assembly

DRL-07

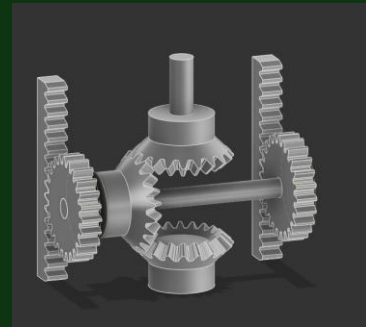
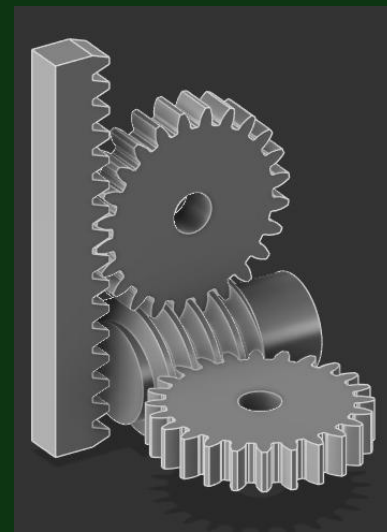
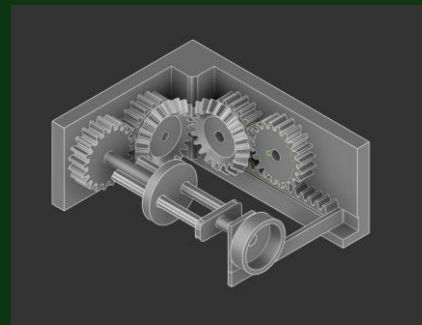
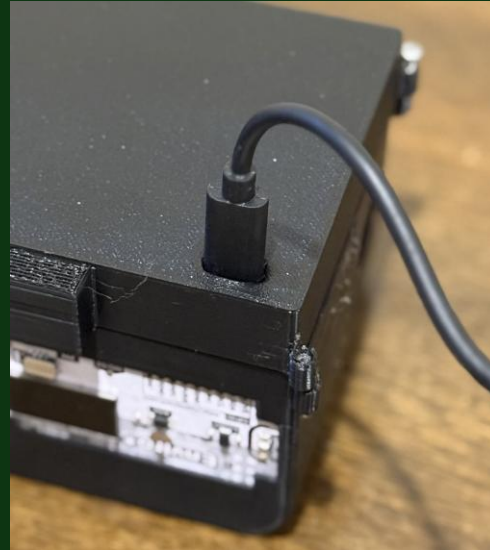
Check structural
integrity of drill
mounting under
torque

DRL-06

Measure power
draw under load to
confirm safe
operation

Test Report Overview (DM)

Test Results



GROUP 24 - UAVPayloadTAQ

RILEY ANDERSSEN

**4th Year Bachelor of Engineering
Software Engineering &
Computer Science**

Target Acquisition & Image Processing



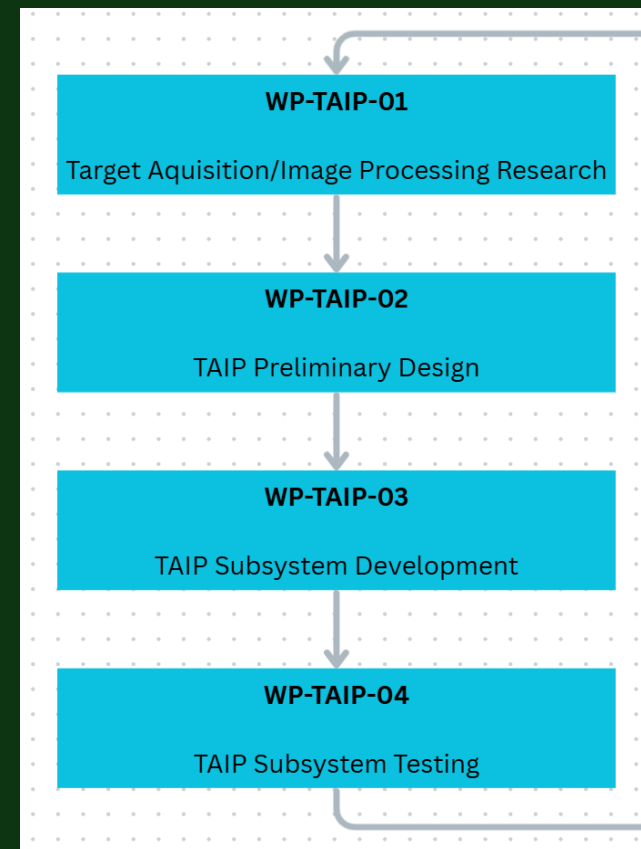
OUR TEAM

Subsystem Overview

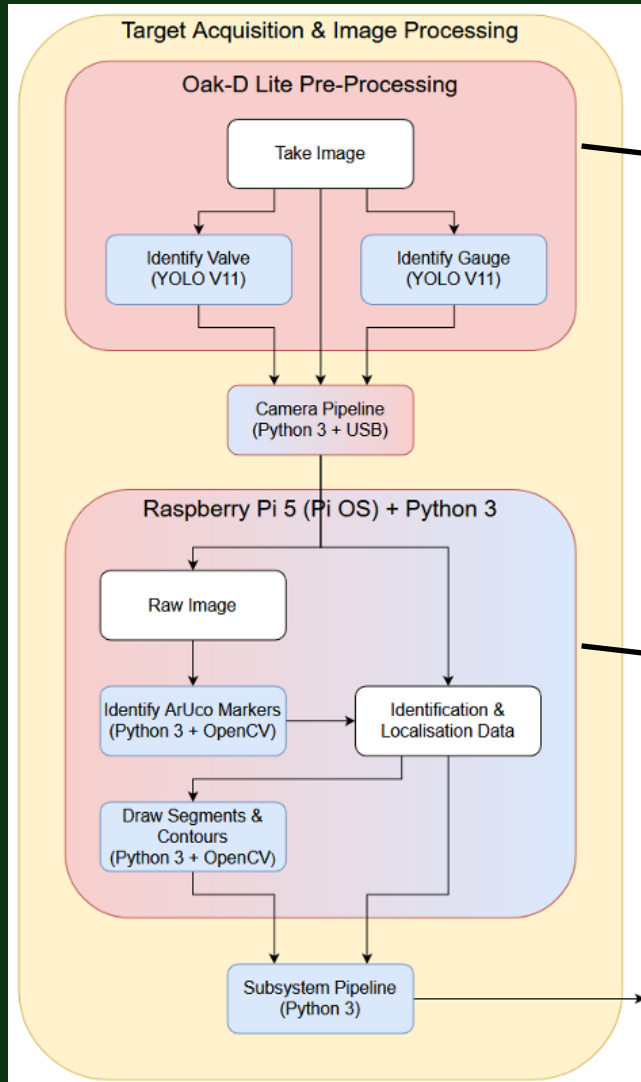
Requirements

- REQ-F-04
- REQ-F-05
- REQ-F-07 – Relies on TAIP feed
- REQ-U-12 – Relies on TAIP feed
- REQ-U-14
- REQ-M-16
- REQ-M-17 – Impacted by TAIP speed
- REQ-M-18 – Impacted by TAIP speed
- REQ-M-19

Work Package



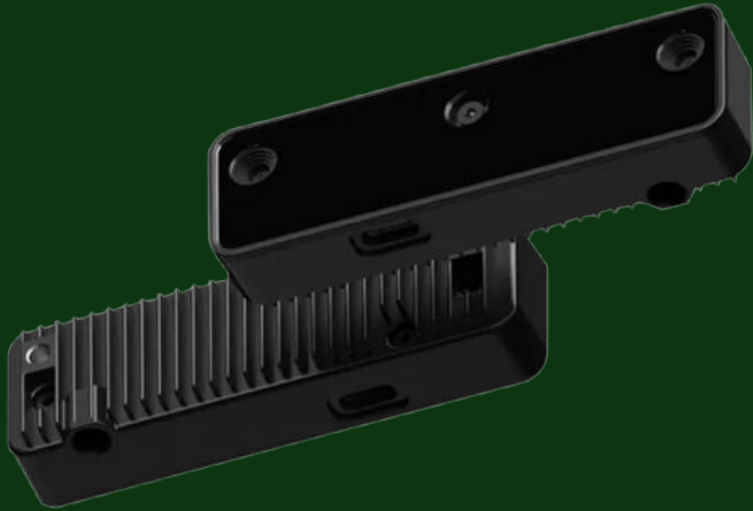
Subsystem Architecture



- Valve and Gauge Detection

- Aruco Marker and Pose Estimation
- Bounding drawing of classifications

Hardware Design Features



Oak-D Lite Camera

- Capture raw images
- On-board RVC2 core
- Operates Yolo V11 nano seg model
- Target detection



Raspberry Pi 5

- Operates main vision program
- Tensor (detection) processing
- Image processing
- Gauge reading calculation

Software Design Features



Python

- Backbone of the subsystem
- Coordinates communication between camera and Pi
- Processes all detections from tensors
- Updates raw image with contours
- Facilitates communication with web UI



OpenCV

- Python library for Computer vision
- Image manipulation
- Detection mask manipulation
- ArUCO detection



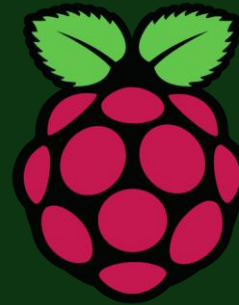
Ultralytics & Yolo

- Yolo V11 nano Segmentation model
- NN model training
- Dataset augmentation



Roboflow

- Dataset generation and splitting
- Image augmentation



Raspberry PiOS

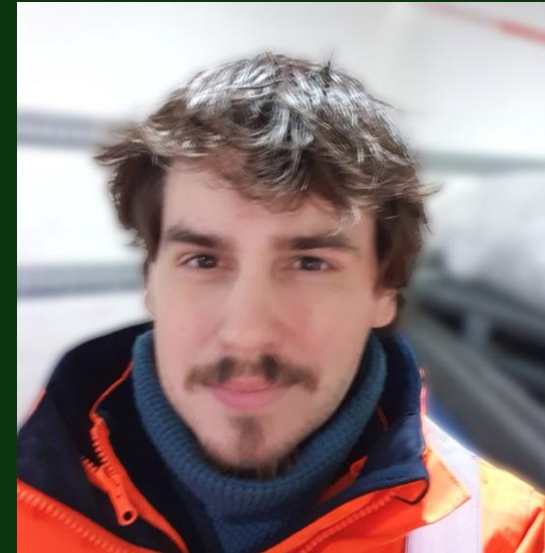
- Main payload operating system

GROUP 24 - UAVPayloadTAQ

NICHOLAS SCHOER

4th Year Bachelor of
Engineering Mechatronics

Target Acquisition & Image Processing



OUR TEAM

Dataset Setup

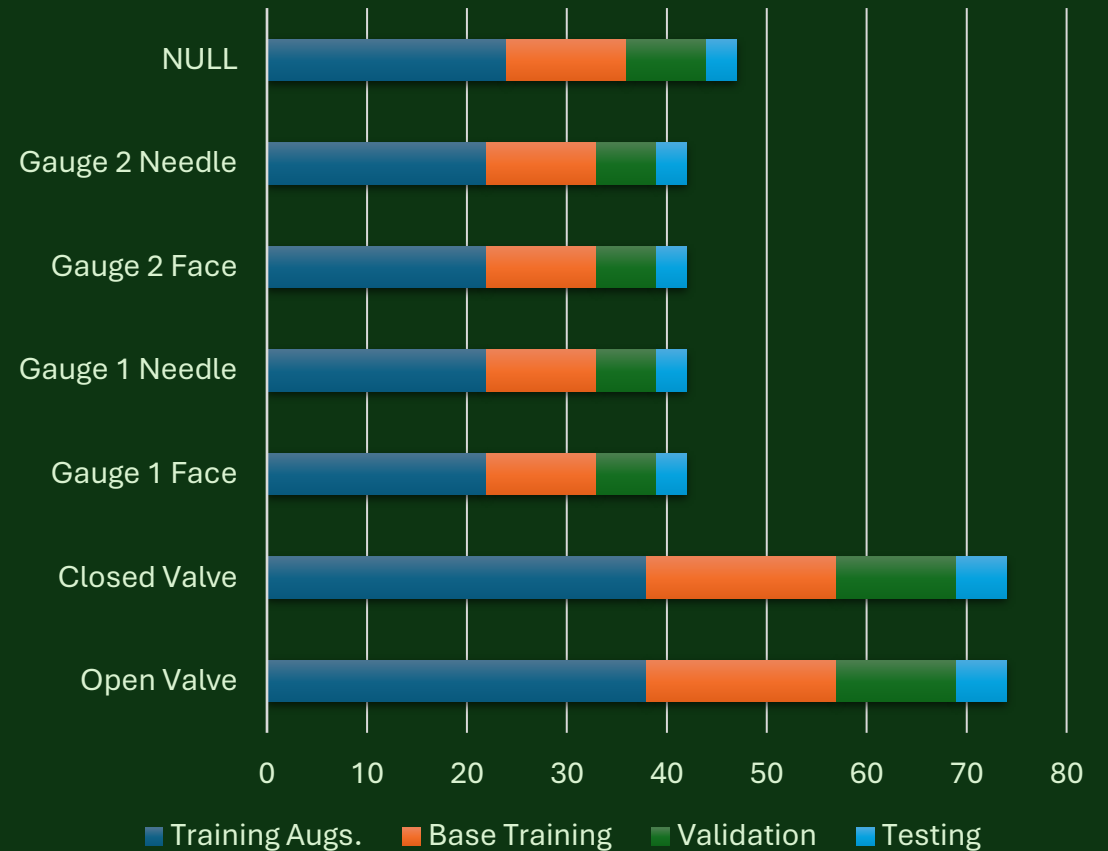
Dataset Wrangling - Roboflow

- Dataset split - 54:32:13
- Light augmentation on training data
- Cropping, rotation, shear, brightness, blur, scale

Model Parameters

- Image size – Training: 864 – Operation: 640
- Shaves: 6

Dataset and Class split



Model Training

Model Parameters

- Training Imgsiz: 864
- Operation Imgsiz : 640
- Shaves: 6

Coarse Training

- Epochs: 500
- Patience: 100
- Further augmentation
- Mosaics

Fine Training

- Epochs: 100
- Patience: 50
- No extra augmentation
- No mosaics



Detection Test Procedures

Physical Testing

- Indoors
- Image displayed on secondary screen
- Varying lighting conditions
- Varying target distances – 0.3m to 2m
- Varying target sizes

Programmatic Tuning

- Adjusting confidence thresholds
- Gating detections



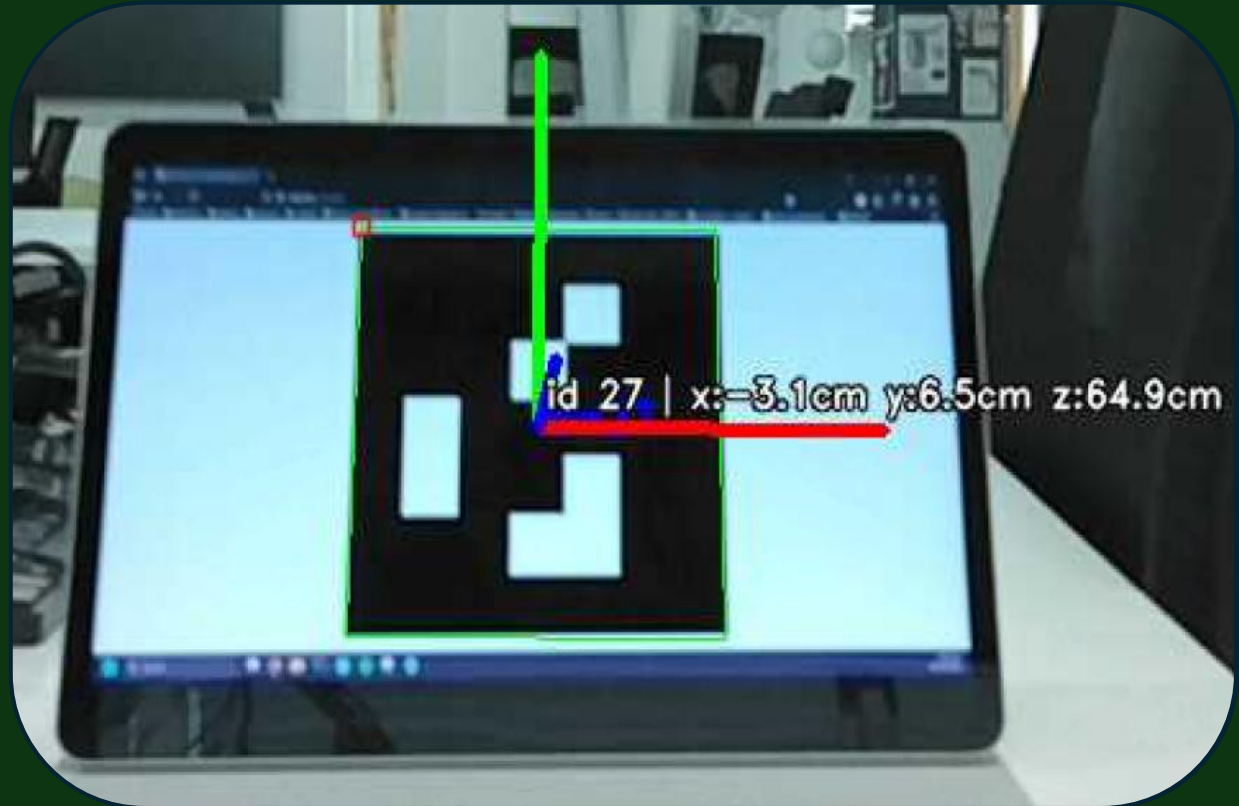
Test Results – Gauges & ArUCO

Gauge Face and Needle

- Range: 0.3m to 1.5m
- No false positives
- Accurate segmentation
- ± 0.2 bar accuracy head-on
- ± 0.5 bar accuracy $> 30^\circ$

ArUCO

- Perfect detection
- Range: 0.3m to +2m
- No false positives
- ± 20 mm pose accuracy



Test Results – Valves

Open Valve

- Range: 0.8m to +2m
- High confidence on dataset
- Low confidence in practice
- False positives

Open Valve

- Range: 0.8m to +2m
- High confidence on dataset
- Low confidence in practice
- No false positives

Better dataset and more training required



GROUP 24 - UAVPayloadTAQ

Rory Blair

**6th Year Bachelor of Engineering
(Computer and Software
Systems)**

Web Visualisation & Interface

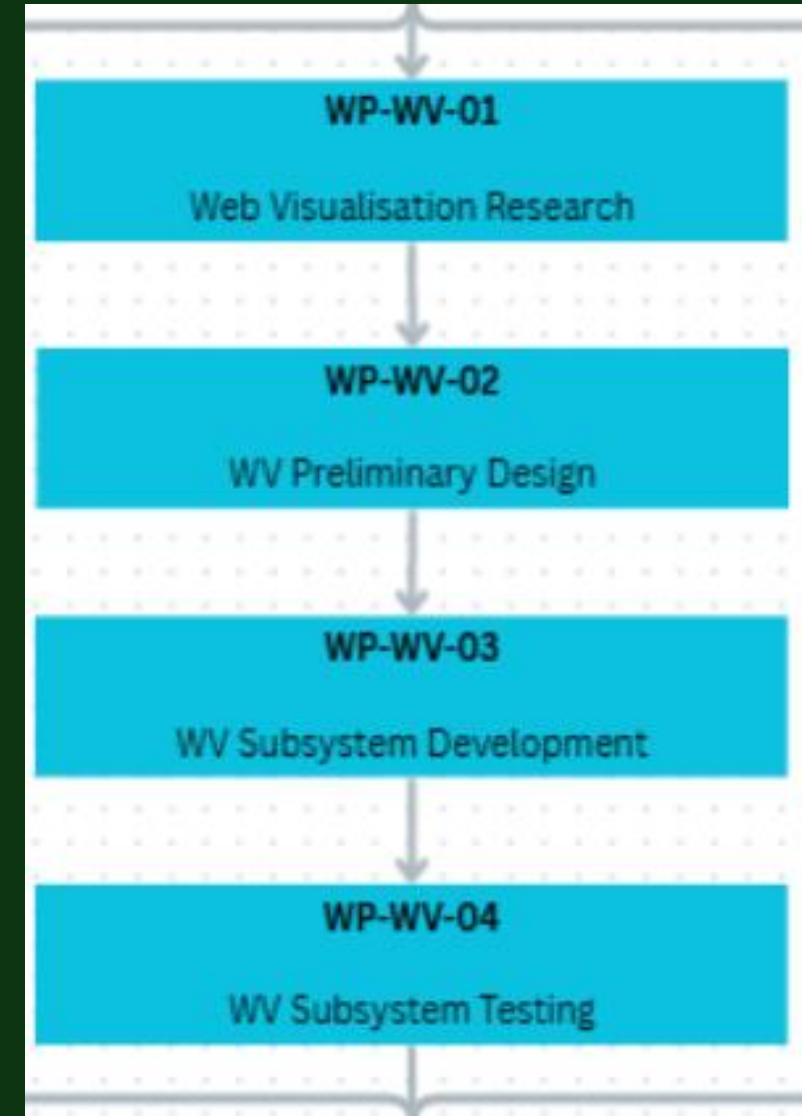


OUR TEAM

Overview

System Requirements:

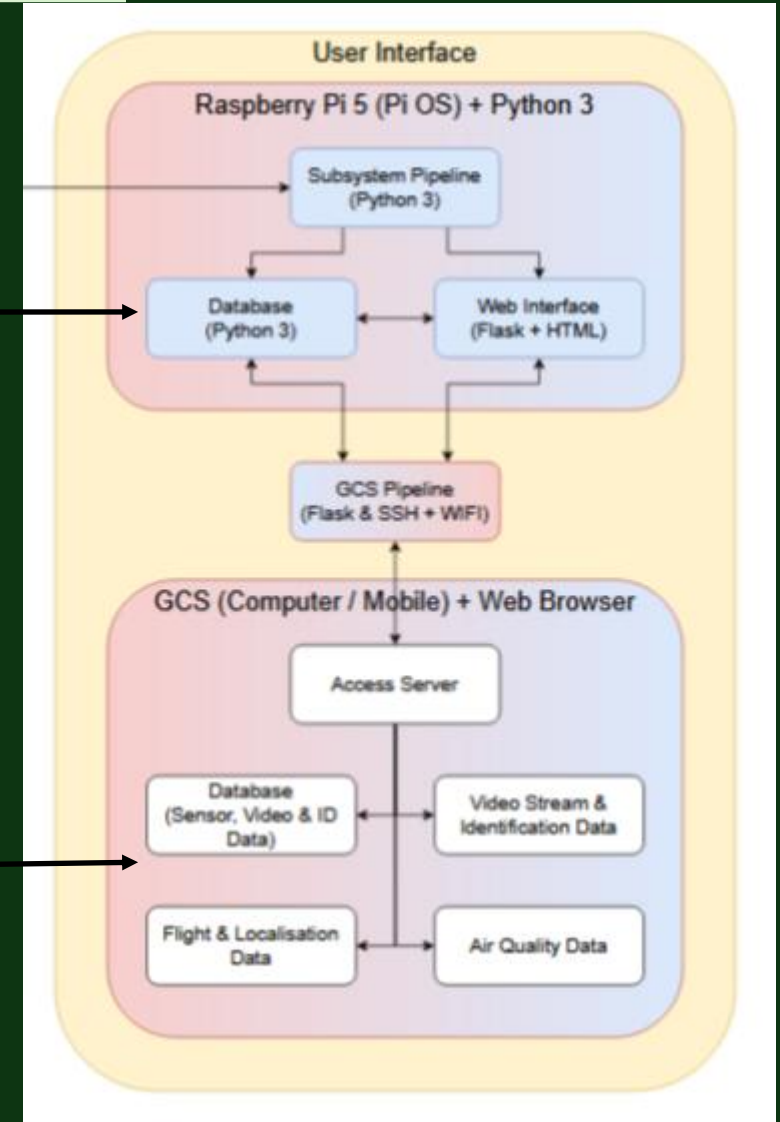
- **REQ-F-04:** Subsystem Communications
- **REQ-F-05:** Alerting System (Dependent on TAIP)
- **REQ-F-06:** Dynamic Real-time Sensors (Dependent on AQ)
- **REQ-F-07:** data transmission and visualisation(dependent on AQ & TAIP)
- **REQ-F-08:** Web Server & log data
- **REQ-U-14:** System Engineering Approach
- **REQ-M-15:** 10-minute logged data
- **REQ-M-16:** Process via Raspberry PI
- **REQ-M-19:** 4-second live feed update



Subsystem Architecture

Back-end

Front-end



Hardware Design Features

Hardware: Raspberry Pi 5

- **Processor:** Broadcom BCM2712 2.4GHz quad-core 64-bit Arm Cortex-A76 CPU
- **GPU:** VideoCore VII GPU
- **Memory:** LPDDR4X-4267 SDRAM
- **GPIO:** 40-pin GPIO header for peripherals and HATs
- **Wi-Fi:** Dual-band 802.11ac (2.4 GHz and 5.0 GHz)
- **Clock:** Onboard RTC with external battery support
- **Power Supply:** 5V/5A USB-C Power Delivery
- **OS:** Raspberry Pi OS (Linux-based)



Software Design Features

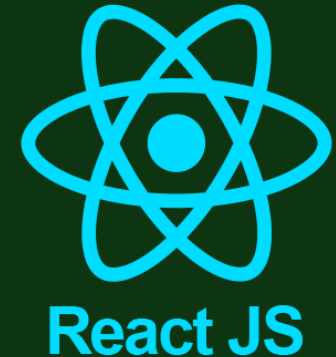
Languages:

- Python: API Routes
- JavaScript: Fetch and Update
- HTML: Website Mark-Up



Frameworks:

- Flask: Back-end
- React: Front-end



Database:

- SQLite3

Web Server:

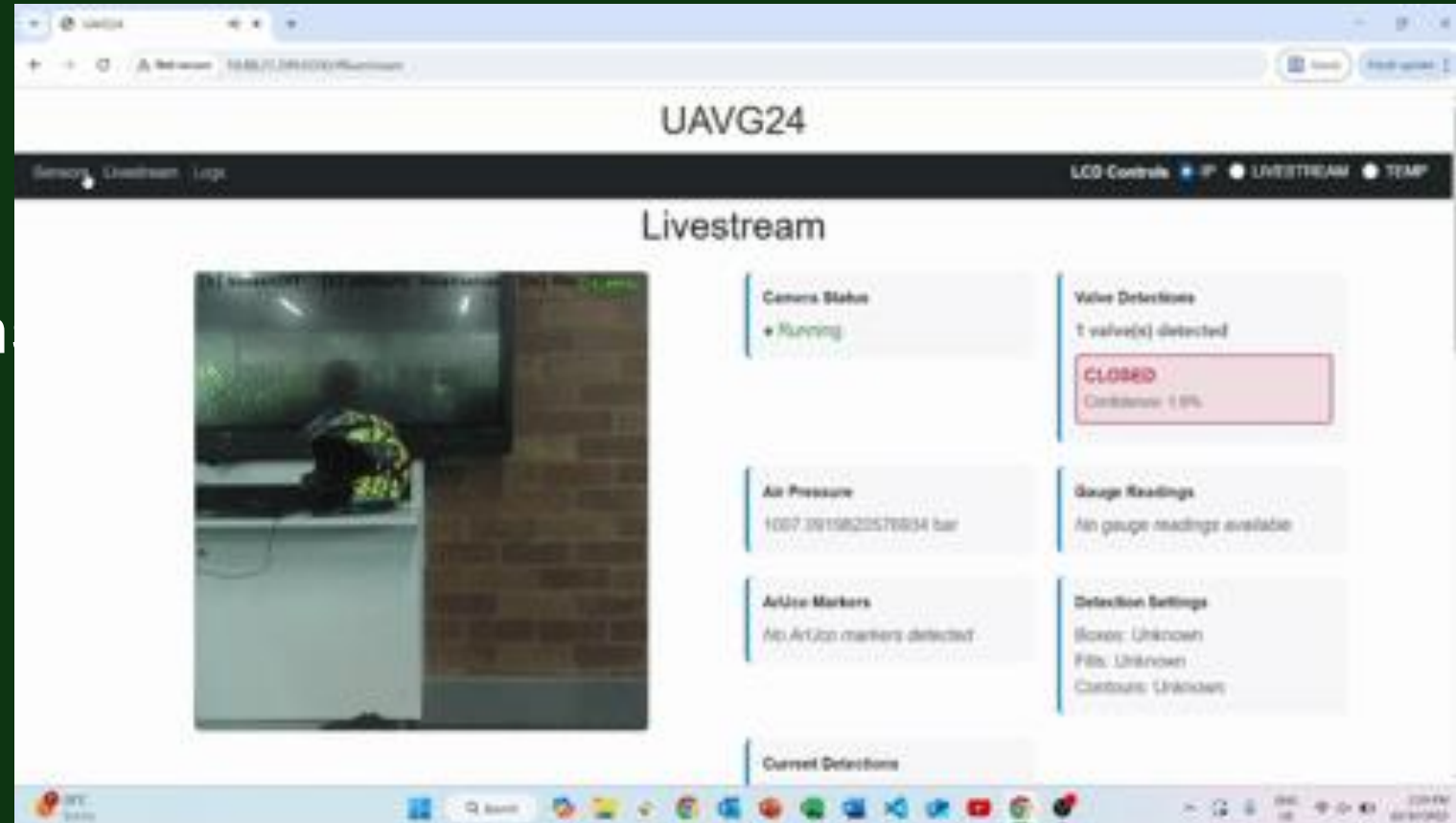
- Threading



Final Design

Test Procedures

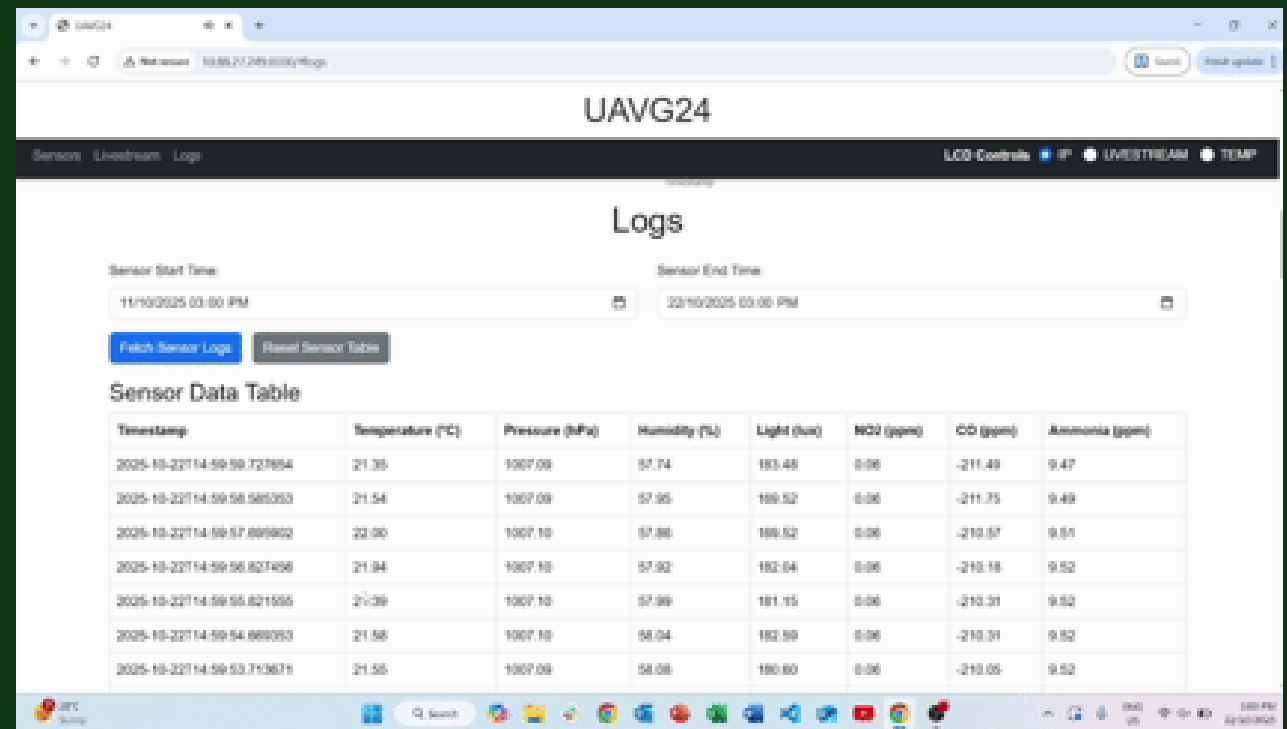
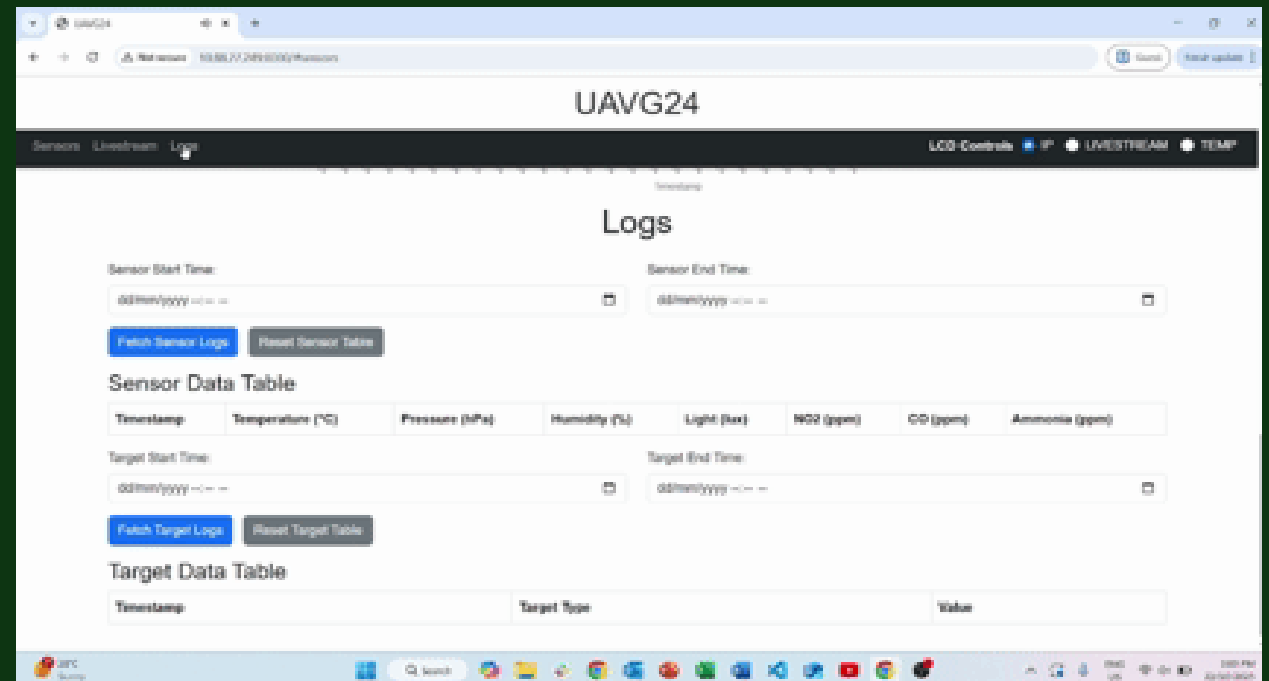
- Creating the web page
- Creating Database
- Test logging and real-time sensor data works with placeholder data



Final Design

Test Results

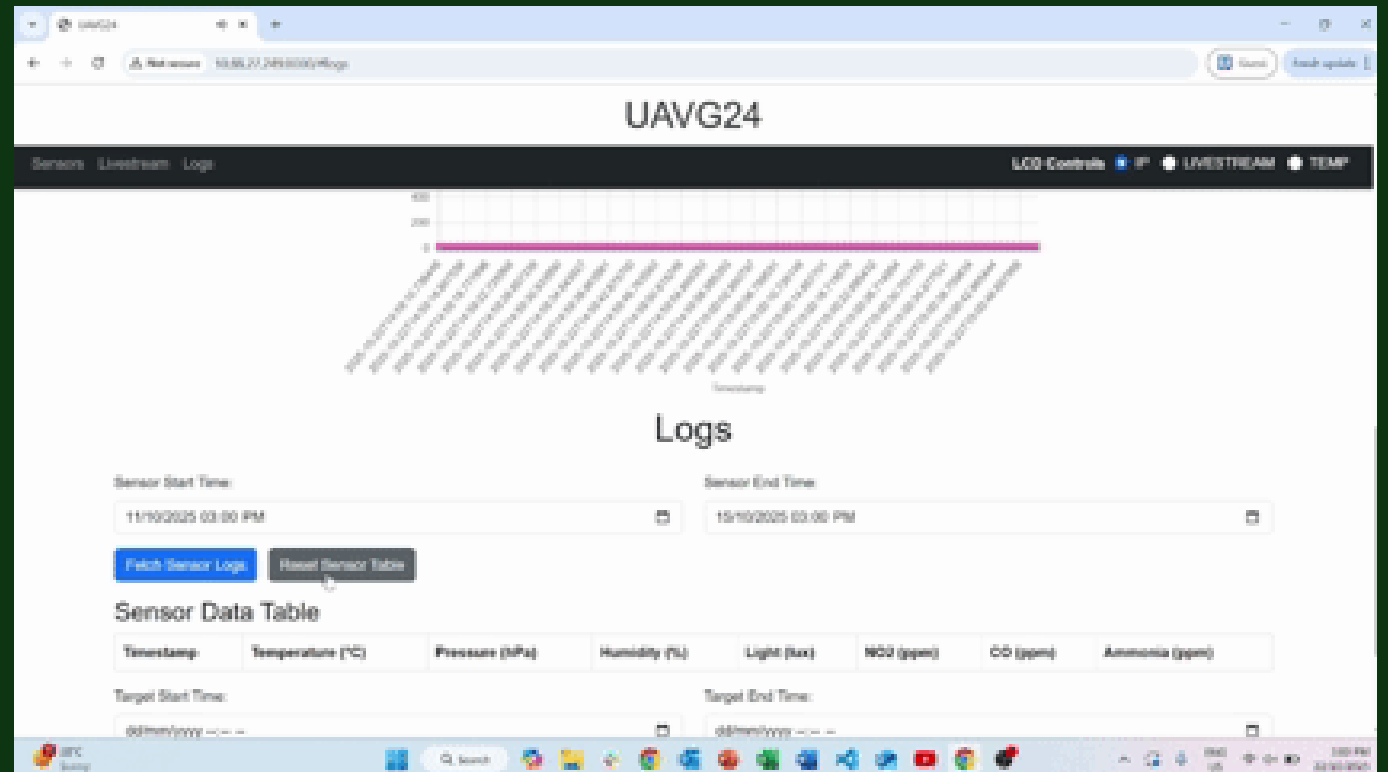
- Sensor Data logs successfully
- Sensor parameter dynamically change on the page.
- Chart visualises:
 - Temperature
 - Pressure
 - Humidity
 - Light
 - N02
 - CO
 - Ammonia



Final Design

Test Results

- Live stream is shown
- Target data is logged



GROUP 24 - UAVPayloadTAQ

ZACH BARLOW

**4th Year Bachelor of Engineering
(Computer and Software
Systems)**

Air Quality and Project Management

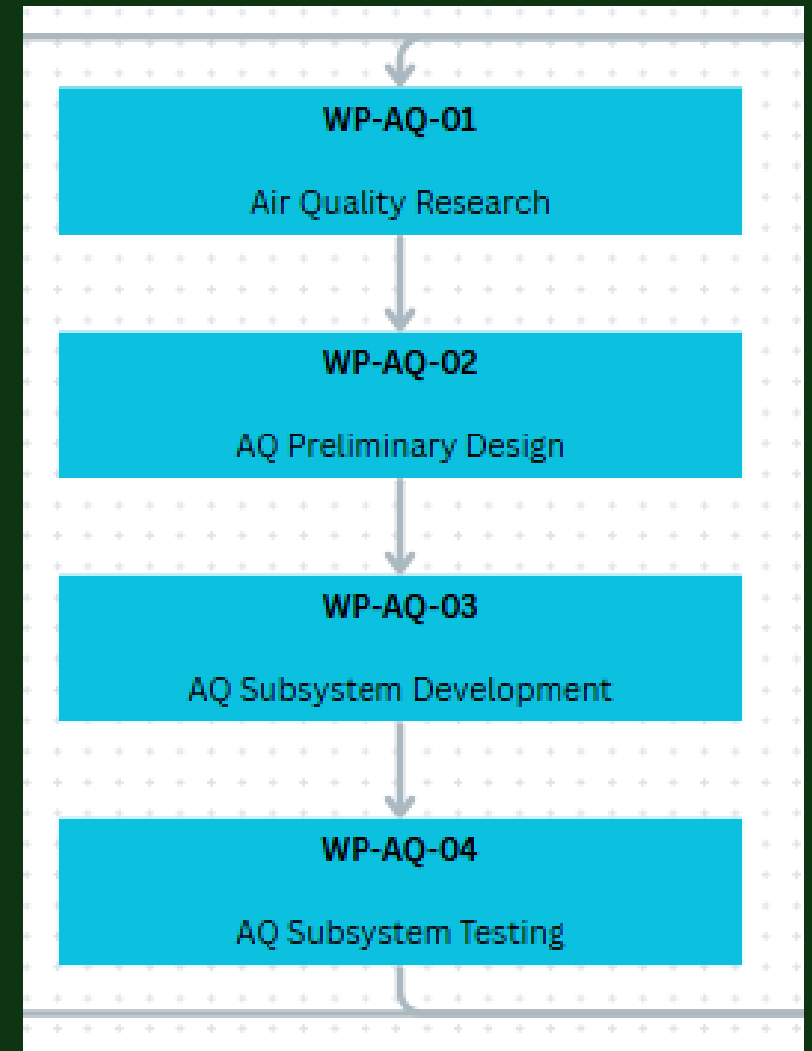


OUR TEAM

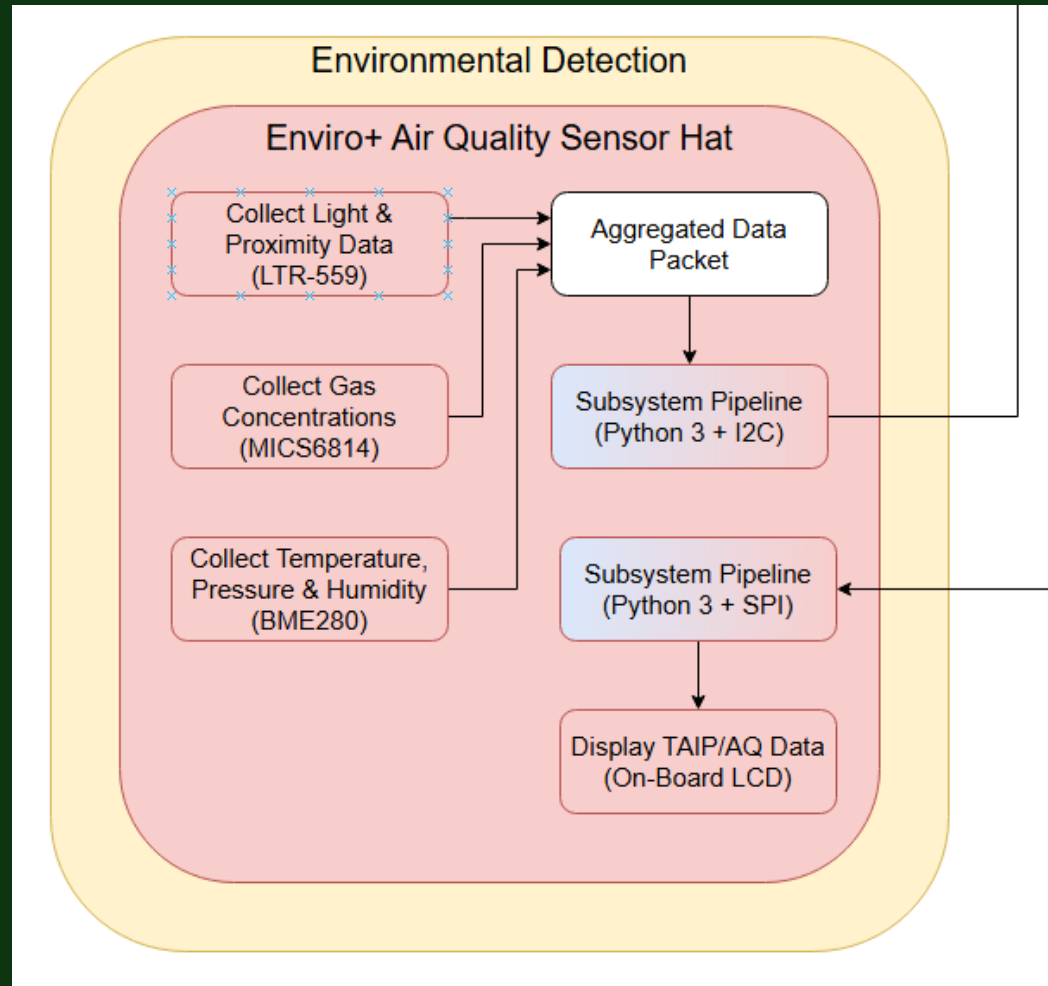
Overview

System Requirements:

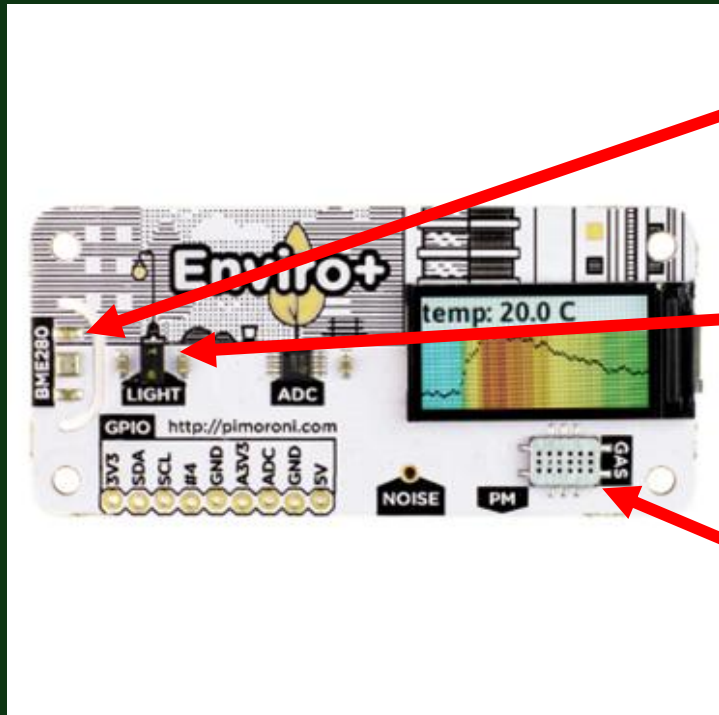
- REQ-F-03
- REQ-F-04
- REQ-U-11
- REQ-U-12
- REQ-M-17
- REQ-M-18
- REQ-M-19



Architecture



Final Design



BME280 Temperature, Pressure and Humidity Sensor

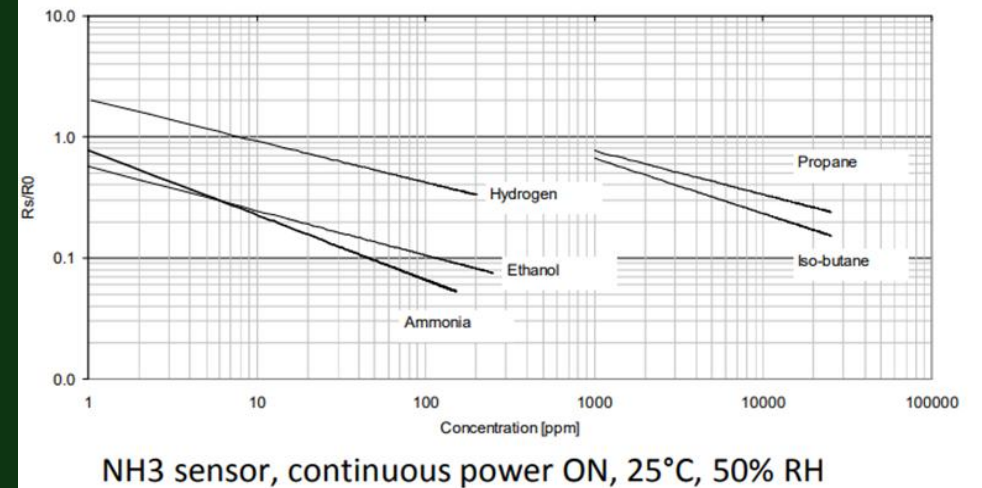
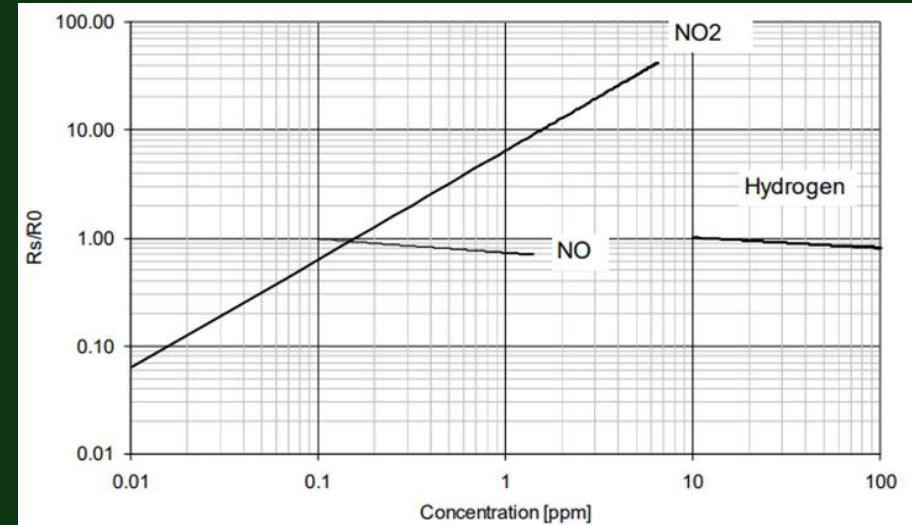
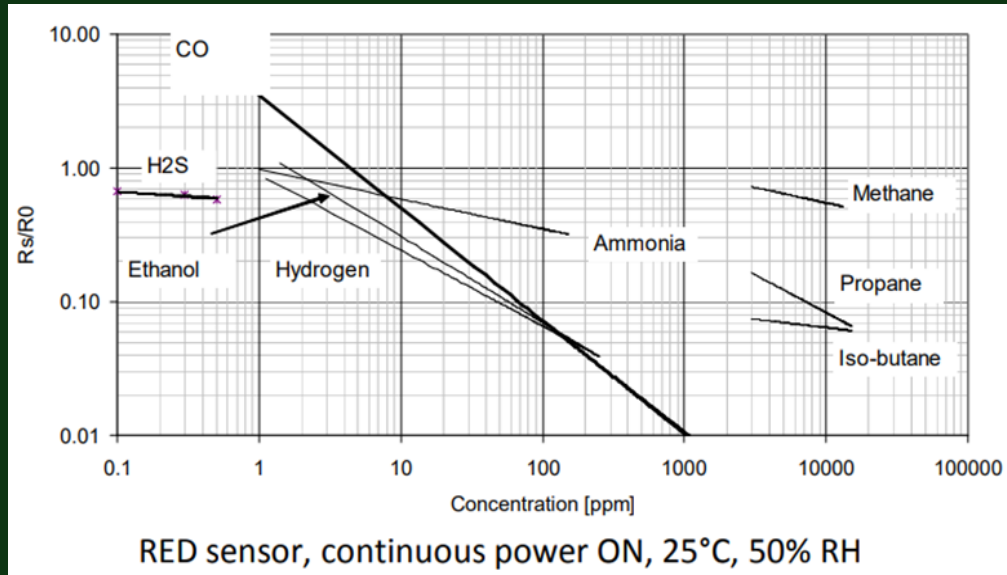


LTR559 Light and Proximity Sensor



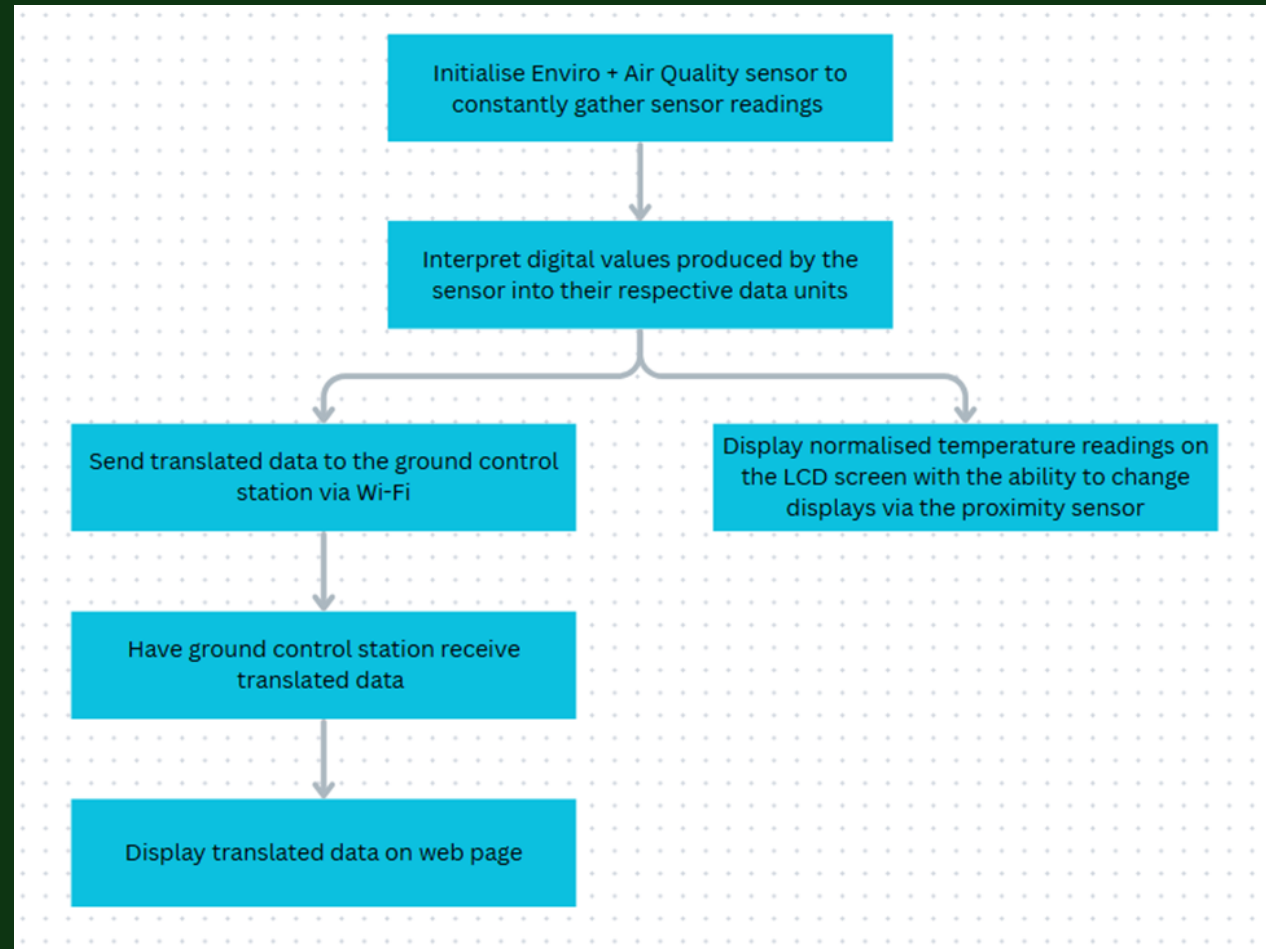
MICS6814 Gas Sensor

Final Design



Final Design

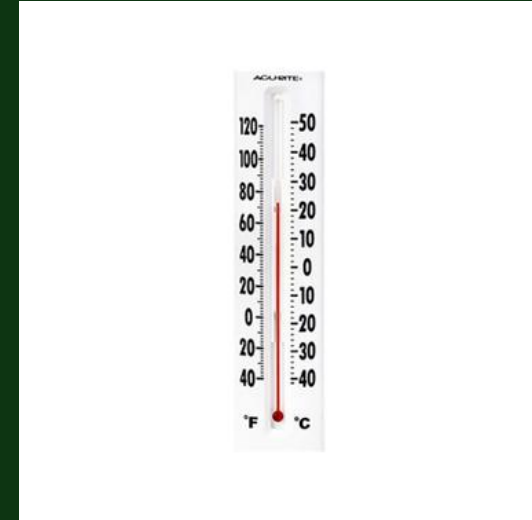
Software Flow Diagram



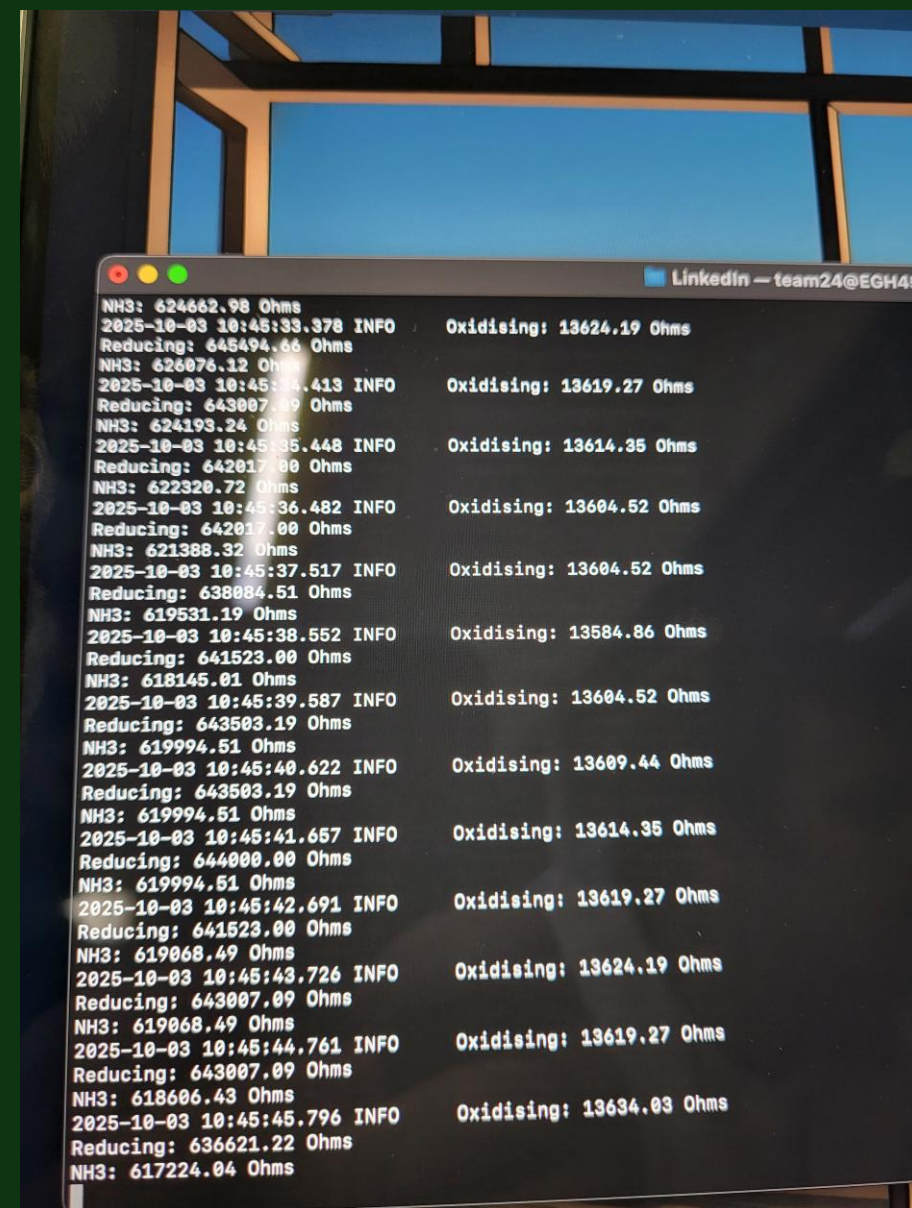
Testing

General Test Procedure:

1. Setup Environment
2. Install Enviro + Library
3. Run example code



Testing



Conclusion

